RedstoneLab

Meters and piston previews for redstone with exact timings

Minecraft 1.21.11 · Fabric · version 0.3.0

by **kyoubitt**



1.1 Installing

- Install **Fabric Loader** for 1.21.11, then **Fabric API**.
- Download `redstonelab-0.3.0.jar` from the GitHub releases page (*not* the `-sources` one).
- Put both jars in your `mods` folder. Nothing else is needed (Mod Menu is optional).
- If both client and server have it, they need the same version.

> Download:
> `github.com/kyoubitt/RedstoneLab/releases`

Installed on	What you get
Your client only	Piston previews on **any** server
Client + server (or singleplayer)	Everything: meters, timeline, exact ticks
Server only	Players without the mod get meters as boss bars and chat, and piston previews

1.2 Hotkeys

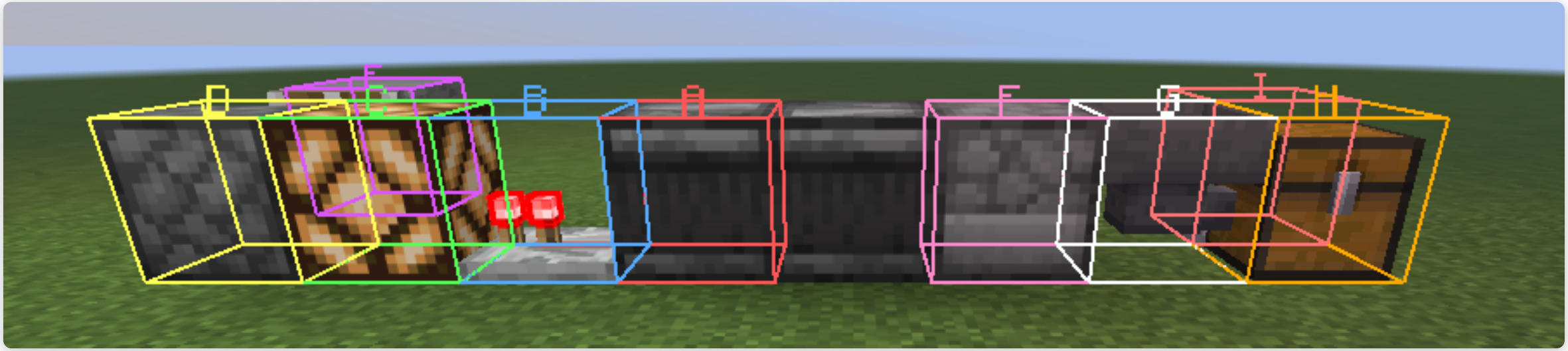
`M`	Meter on / off (block you look at)
`Backspace`	Remove a meter
`R` + `M`	Remove all meters
`R` + `Space`	Freeze / unfreeze the timeline
`R` + `Left`	1 tick back
`R` + `Right`	1 tick forward
`P`	Hide piston previews
`R` + `G`	Meter groups
`R` + `C`	Settings



> **R + Left** means hold R, then press Left. Change any of them in **Settings** → **Hotkeys**: click one, hold the new keys, let go. Esc cancels.

2.1 Adding meters

- Look at a block and press `M`. Press `M` again to remove it.
- `Backspace` removes the meter you look at (or the newest one). `R` + `M` removes all.
- Any block works: dust, repeaters, comparators, torches, lamps, observers, pistons, droppers, hoppers, chests…
- Each meter gets a letter and a coloured outline. Meters move along with blocks pushed by pistons.



2.2 Reading the timeline



1. **Order**: which meters turn on first, counted from A. `A → 2t → B D F G` = B, D, F and G turn on 2 ticks after A.
2. The meter's letter and colour.
3. **Bar** = on. The number = how many ticks it lasted. One column = one tick.
4. What the block is and its on / off lengths.

`6t` always 6 **`3t/5t`** alternates **`2t/3t/3t`** repeating pattern **`2t or 3t`** no pattern **`thin line`** powered, not on yet **`white line`** 0-tick pulse

2.3 What each meter tells you

Block	Bar means	Text
Dust, repeater, torch, lamp, observer	on	`on 2t · off 4t`
Piston	out	`moves every 6t · out 2t`
Dropper, dispenser, crafter	fires	`27 Stone · fires every 4t`
Hopper	locked	items, `out every 8t`
Chest, barrel…	comparator signal above 0	items, `signal 1 (2 at 124)` = the comparator gives 2 at 124 items
Comparator	on	`subtract · back 15 · side 14 · out 1`

2.4 Sub-tick order



1. `R` + `Left` goes 1 tick back and freezes the timeline. `R` + `Right` goes forward; back at now it runs again.
2. Everything that happened at your meters in that tick, **in the order the game ran it**, grouped by the part of the tick.

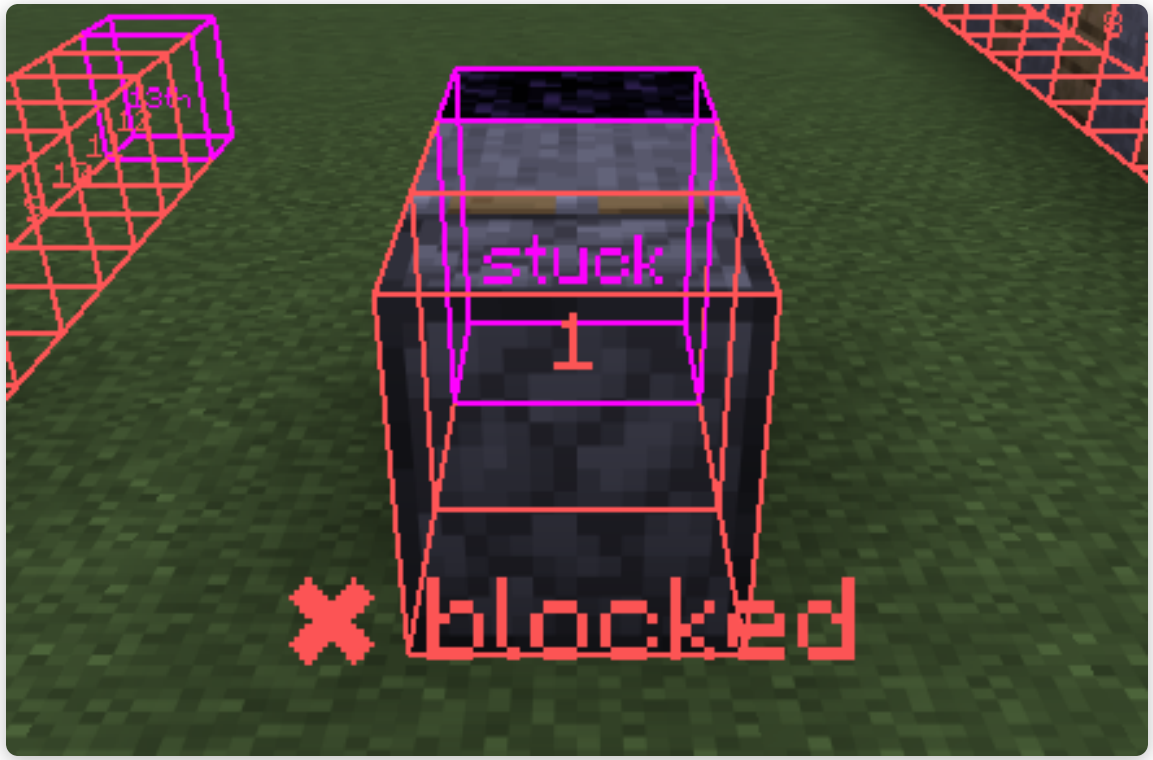
> Want block and shape updates too? Settings → **Show event marks**.

2.5 Checking and organising

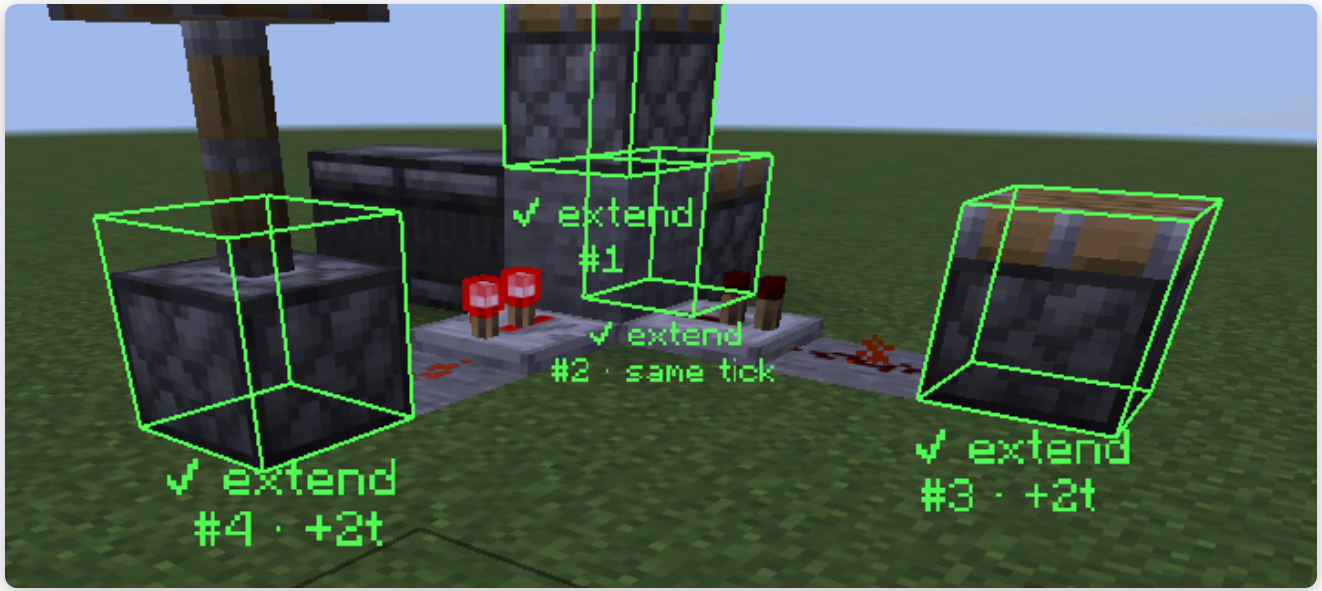
`/rlab meter expect A 2 4`	A should be on 2t, off 4t. Wrong pulses turn red and you get told.
`/rlab meter name A clock`	A name in front of A's text.
`/rlab group <name>`	Separate saved meter sets. `R` + `G` lists them (click one to switch).
`/rlab meter`	The same info in chat (works without the client mod, plus boss bars).
`/rlab tool`	A tool: right-click blocks to add meters (no hotkeys needed).

3.1 Piston previews

- **Right-click a piston with an empty hand**: it shows what it will do next. Click again to hide it.
- Show several at once. `P` hides them all. Sneaking does nothing.
- ✔ **push 3**: numbers on the blocks = the order they move. `breaks` = destroyed.
- ✖ **blocked**: `stuck` marks the block in the way.
- ✖ **too many** (over 12): `13th` marks the extra block.
- ⚡ **QC** = powered only through the block above.
  ⚡ **BUD** = moves on the next block update.
- Works on **any server**: only your client needs the mod.



3.2 Firing order



- Show several pistons: each gets its place in the order they extend, counted from the first one you showed.
- `#2 · +2t` = 2 ticks after the one before.
- `same tick` = same tick, but the game moved it after the one before.
- `own timing` = runs at a different speed than #1.

> Exact with RedstoneLab on the server (or singleplayer). On other servers the order is still exact; the ticks between can be off by one when the server lags.

4 Settings

- Open with `R` + `C` (or Mod Menu).
- **Meters**: what the timeline shows, how many ticks, event marks.
- **GUI**: where it sits, size, background, which parts show.
- **Pistorder**: also outline where pushed blocks land.
- **Hotkeys**: every key, combos allowed.
- Every change saves right away. **Reset** brings back the default. Hover a setting to see what it does.



That's it

Add a few meters, freeze the timeline, show a piston: go build something.

Download & source	`github.com/kyoubitt/RedstoneLab`
Commands	`/rlab help`
License	LGPL-3.0